COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	7/8/2026	**DATE SUBMITTED:**	7/10/2026
NAME:	Ralph Angelov F. Braganza	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. _1.1__ Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

Seat work 1

Perform the following:
Create an Algorithm, Pseudocode, Flowchart, and C++ Program for the following:

1. Compute the total and average of 3 quizzes then display the total and average.

ALGORITHM

1. Start
2. Create two variables:
   - One to store the total quiz score.
   - One to receive the quiz score entered by the user.
3. Ask the user to enter a quiz score.
4. Check if the input is a valid score (0–100).
   - If the input is invalid, display an error message and go back to Step 3.
5. Add the valid quiz score to the total quiz score.
6. Repeat Steps 3–5 until 3 valid quiz scores have been entered.
7. Create a variable to store the average of the 3 quiz scores.
   - average = total quiz score / 3`
8. Display:
   - Quiz Average
   - Total Quiz Score
9. End

PSEUDOCODE:

START

```
SET quiz_total = 0 as FLOAT
SET qInput as FLOAT

FOR i <= 3

  REPEAT
    DISPLAY "Enter Quiz Score ", quiz, " (0-100): "
    INPUT qInput
```

```
        IF input is invalid THEN
            DISPLAY "Invalid input. Please enter an integer."
        ELSE IF qInput < 0 OR qInput > 100 THEN
            DISPLAY "Invalid score. Must be between 0 and 100."
        END IF

    UNTIL input is valid AND qInput is between 0 and 100

    quiz_total = quiz_total + qInput

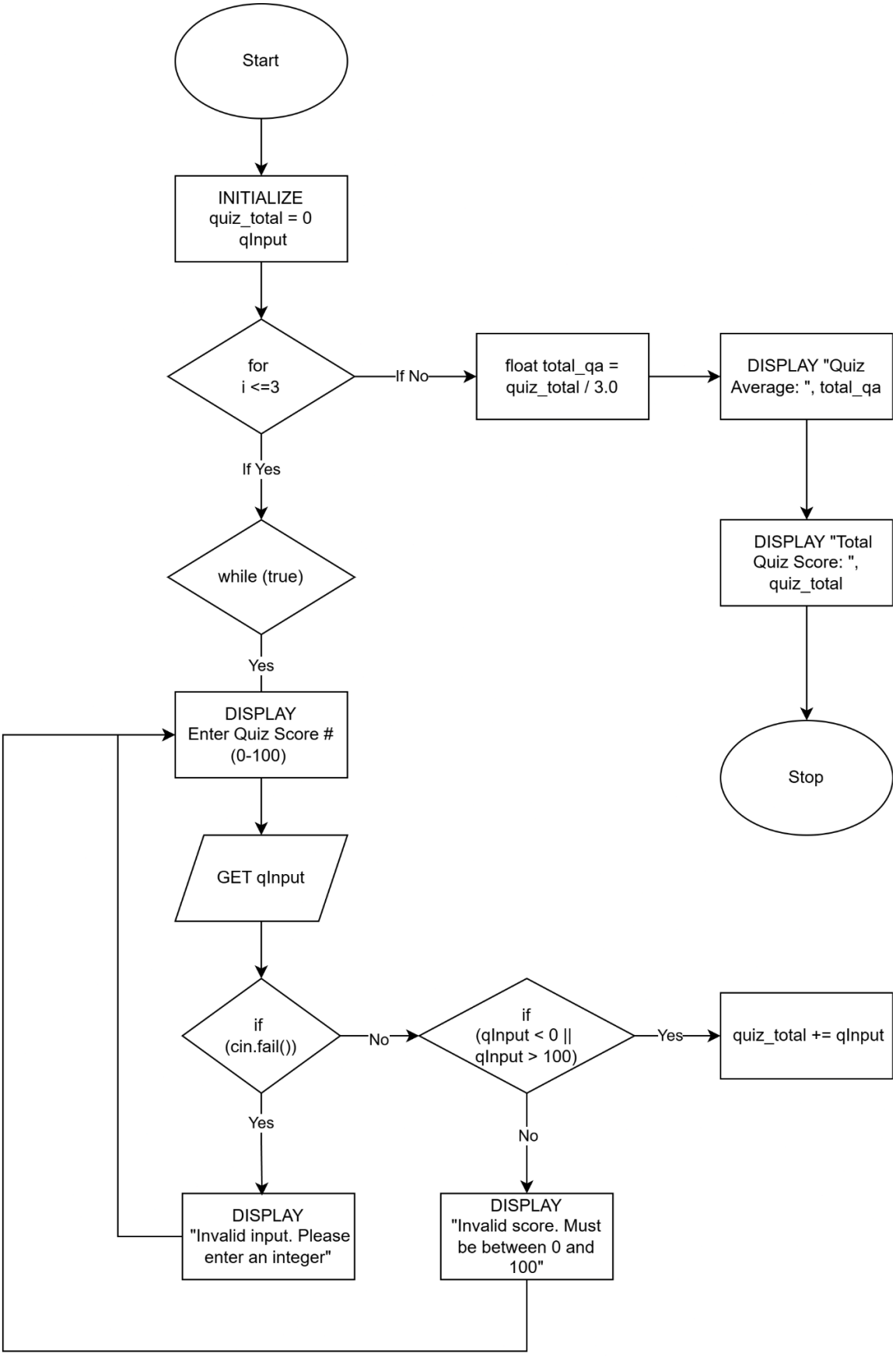
END FOR

SET average = quiz_total / 3

DISPLAY "Quiz Average: ", average
DISPLAY "Total Quiz Score: ", quiz_total

STOP
```

FLOWCHART:


Start
INITIALIZE
quiz_total = 0
qInput
for
i <=3
If No
float total_qa =
quiz_total / 3.0
DISPLAY "Quiz
Average: ", total_qa
If Yes
DISPLAY "Total
Quiz Score: ",
quiz_total
while (true)
Yes
DISPLAY
Enter Quiz Score #
(0-100)
Stop
GET qInput
if
(cin.fail())
No
if
(qInput < 0 ||
qInput > 100)
Yes
quiz_total += qInput
Yes
No
DISPLAY
"Invalid input. Please
enter an integer"
DISPLAY
"Invalid score. Must
be between 0 and
100"

CODE AND OUTPUT:

```cpp
#include <iostream>
#include <limits>
using namespace std;

int main() {
    float quiz_total = 0;
    float qInput;

    /* This block of code will ask the user for the 3 quiz scores. If the user fails to input a
       number from 0 to 100, it will tell the user to try again until they input the right values.*/
    for (int i = 1; i <= 3; i++) {
        while (true) {
            cout << "Enter Quiz Score " << i << " (0-100): ";
            cin >> qInput;

            if (cin.fail()) {
                cout << "Invalid input. Please enter an integer.\n";
                cin.clear();
                cin.ignore(numeric_limits<streamsize>::max(), '\n');
                continue;
            }

            if (qInput < 0 || qInput > 100) {
                cout << "Invalid score. Must be between 0 and 100.\n";
                continue;
            }

            break;
        }
        quiz_total += qInput;
    }

    // This line of code will display the average quiz score and total quiz score.
    float total_qa = quiz_total / 3.0;
    cout << "\nQuiz Average: " << total_qa << "\n";
    cout << "Total Quiz Score: " << quiz_total << "\n";

    return 0;
}
```

```
Enter Quiz Score 1 (0-100): 99
Enter Quiz Score 2 (0-100): eighty eight
Invalid input. Please enter an integer.
Enter Quiz Score 2 (0-100): 88
Enter Quiz Score 3 (0-100): 100

Quiz Average: 95.6667
Total Quiz Score: 287
```

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes

change. Display the transactions made.

ALGORITHM:

1. Start

2. Create variables to store:
   - The names of the 3 items.
   - The prices of the 3 items.
   - Total amount.
   - Payment amount.
3. Ask the user to enter the name of the item.
4. Ask the user to enter the price of the item.
5. Check if the entered price is a valid number.
   - If the input is invalid, display an error message and go back to Step 4

6. Store the valid price in its corresponding variable.
7. Add the valid price to the total amount.
8. Display the total amount of all items.
9. Ask the user to enter the payment amount.
10. Check if the payment is valid.
    - If the payment is not a valid number go back to step 9:
11. Calculate the change by subtracting the total amount from the payment.
12. Display the receipt:
    - Item names
    - Item prices
    - Total amount
    - Payment amount
    - Change

13. End

PSEUDOCODE:

```
Start

   SET item1, item2, item3 as STRING
   SET price1, price2, price3 as FLOAT
   SET total, payment, price as FLOAT

   SET total = 0

   FOR i <= 3

      IF i = 1
         DISPLAY "Enter Item 1 Name: "
         INPUT  item1
      ELSE IF i = 2
         DISPLAY "Enter Item 2 Name: "
```

```
        INPUT  item2
    ELSE
        DISPLAY "Enter Item 3 Name: "
        INPUT item3
    END IF

    REPEAT
        DISPLAY "Enter Item ", i, " Price: "
        INPUT price

        IF input is invalid
            DISPLAY "Invalid input. Please enter a number."
        ELSE IF price < 0
            DISPLAY "Price cannot be negative."
        END IF

    UNTIL input is valid AND price >= 0

    IF i = 1
        price1 = price
    ELSE IF i = 2
        price2 = price
    ELSE
        price3 = price
    END IF

    total = total + price

END FOR

DISPLAY "Total Amount: ", total

REPEAT
    DISPLAY "Enter Payment: "
    INPUT payment

    IF input is invalid
        DISPLAY "Invalid input. Please enter a number."
    ELSE IF payment < total
        DISPLAY "Insufficient payment. Please enter enough payment."
    END IF

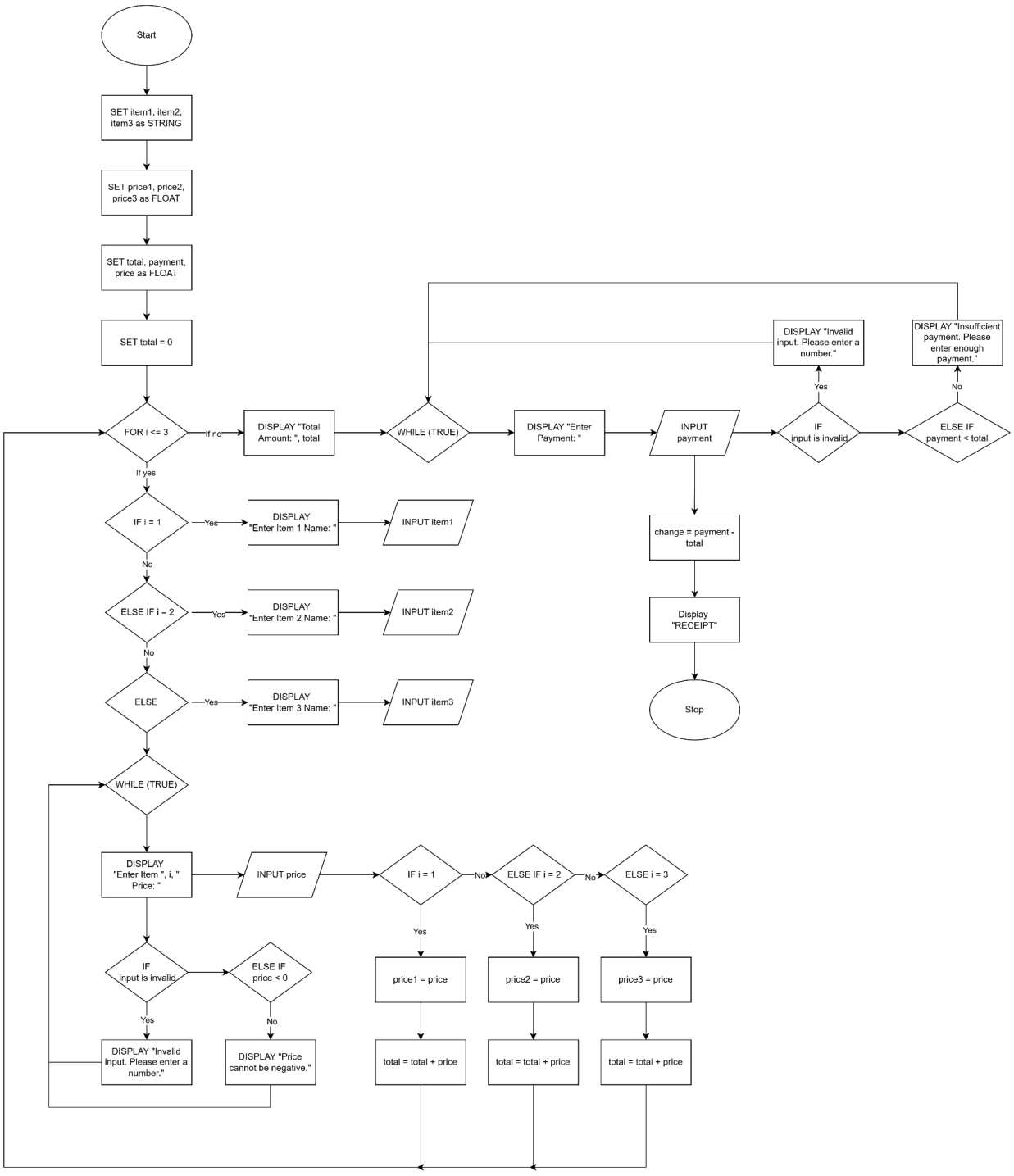
UNTIL input is valid AND payment >= total

change = payment - total

Display "RECEIPT"
```

Stop

FLOWCHART:



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

CODE AND OUTPUT:

```cpp
#include <iostream>
#include <limits>
using namespace std;

int main() {
    string item1, item2, item3;
    float price1, price2, price3;
    float total = 0;
    float payment;

    for (int i = 1; i <= 3; i++) {

        // Item Name - this block of code will go through each item one by one.
        if (i == 1) {
            cout << "Enter Item 1 Name: ";
            cin >> item1;
        }
        else if (i == 2) {
            cout << "Enter Item 2 Name: ";
            cin >> item2;
        }
        else {
            cout << "Enter Item 3 Name: ";
            cin >> item3;
        }

        // Item Price - this while loop will keep running until a valid number is inputted and then add
            up all the prices.
        while (true) {
            cout << "Enter Item " << i << " Price: ";
```

```
30              float price;
31              cin >> price;
32
33              if (cin.fail()) {
34                  cout << "Invalid input. Please enter a number.\n";
35                  cin.clear();
36                  cin.ignore(numeric_limits<streamsize>::max(), '\n');
37                  continue;
38              }
39
40              if (price < 0) {
41                  cout << "Price cannot be negative.\n";
42                  continue;
43              }
44
45              if (i == 1)
46                  price1 = price;
47              else if (i == 2)
48                  price2 = price;
49              else
50                  price3 = price;
51
52              total += price;
53              break;
54          }
55      }
56
57      cout << "\nTotal Amount: " << total << endl;
58
```

```
58
59      // Payment—this is where the payment is done; it will ask the user for the needed amount and it
            will keep asking until a valid input or amount is made.
60      while (true) {
61          cout << "Enter Payment: ";
62          cin >> payment;
63
64          if (cin.fail()) {
65              cout << "Invalid input. Please enter a number.\n";
66              cin.clear();
67              cin.ignore(numeric_limits<streamsize>::max(), '\n');
68              continue;
69          }
70
71          if (payment < total) {
72              cout << "Insufficient payment. Please enter enough payment.\n";
73              continue;
74          }
75
76          break;
77      }
78
79      // Receipt—it will print a receipt for the user for items 1 to 3 and their individual prices. The
            receipt will also print out the total, the amount you paid, and the change.
80      cout << "\n==================================";
81      cout << "\n             RECEIPT";
82      cout << "\n==================================";
83      cout << "\nItem\t\tPrice";
84      cout << "\n----------------------------------";
85      cout << "\n" << item1 << "\t\t" << price1;
86      cout << "\n" << item2 << "\t\t" << price2;
87      cout << "\n" << item3 << "\t\t" << price3;
88      cout << "\n----------------------------------";
```




```
89      cout << "\nTotal:\t\t" << total;
90      cout << "\nPayment:\t" << payment;
91      cout << "\nChange:\t\t" << payment - total;
92      cout << "\n==================================\n";
93
94      return 0;
95  }
```



```
Enter Item 1 Name: Bread
Enter Item 1 Price: 85
Enter Item 2 Name: Apple
Enter Item 2 Price: Juice
Invalid input. Please enter a number.
Enter Item 2 Price: 30
Enter Item 3 Name: Milk
Enter Item 3 Price: 100

Total Amount: 215
Enter Payment: three hundred
Invalid input. Please enter a number.
Enter Payment: 300

==================================
             RECEIPT
==================================
Item        Price
----------------------------------
Bread       85
Apple       30
Milk        100
----------------------------------
Total:      215
Payment:    300
Change:     85
==================================
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT and Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar checking, code assistance, error assistance using VS Code and formatting**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **By making an algorithm and pseudocode first, I had an idea of what I wanted to do for the code; of course, I had to research how to do everything in C++ again due to other courses overriding my knowledge (like OOP 1). I used AI to explain the concepts to me so that I will be able to create my own version without using it, but if I do go through some sort of problem, I will try to solve it by myself, and if that takes me more than 10 minutes, I will ask AI for assistance. This method has helped me not be so reliant on AI, but rather it has helped me utilise its capabilities for learning.**

COMPUTER ENGINEERING

QUEZON CITY

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.